Aditya Bidwai

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Education

University of Minnesota Twin Cities — *Minneapolis, MN, USA*
Master of Science (M.S.), Robotics — *Sep 2024 – Sep 2027*

Birla Institute of Technology and Science, Pilani (BITS-Pilani) — *Goa, India*
Bachelor of Engineering (B.E.), Electronics and Communication — *Aug 2018 – May 2022*

Publications

- [Author 1], [Author 2], [Author 3], **Bidwai, A.**, et al. (2026). *Research on Cooperative Multi-Agent Reinforcement Learning.* **Under Review at AAMAS 2027**.
- Dai, W., **Bidwai, A.**, & Sartoretti, G. (2024). *Dynamic Coalition Formation and Routing for Multirobot Task Allocation via Reinforcement Learning.* Published at **IEEE ICRA 2024**. **(paper)**
- Gong, Y., Sun, G., Nair, A., **Bidwai, A.**, Cs, R., Grezmak, J., ... Daltorio, K. A. (2023). *Legged robots for object manipulation: A review.* Published in **Frontiers in Mechanical Engineering**. **(paper)**
- Dhongdi, S., Tahiliani, M., Mehta, O., Dharmadhikari, M., Agrawal, V., & **Bidwai, A.** (2022). *FANS: flying ad-hoc network simulator.* Published at **2022 ACM LANC** (Latin America Networking Conference). **(paper)**

Research Experience

ACRES Lab, University of Minnesota Twin Cities — *Minneapolis, MN*
Research Assistant (Supervisor: Prof. Peng Wei) — *Oct 2026 – Present*
- Researching robotic manipulation for agricultural tasks such as harvesting and pruning, using active perception to handle visual occlusions; studied imitation learning policies including ACT, Diffusion Policy, and VLA models
- Set up and integrated Seeed Studio and YAM Pro robot arms for manipulation experiments

OptimalX Research Lab, University of Minnesota Twin Cities — *Minneapolis, MN*
Research Assistant (Supervisor: Prof. Yue Yu) — *Jan 2025 – Sep 2026*
- Developed motion planning algorithms for UAV monitoring in energy-constrained environments (wildfires) using Mixed-Integer Linear Programming (MILP) -based optimization (Gurobi) and Markov Decision Process (MDP) methods
- Built an autonomous UAV (PX4, Pixhawk, RPi) for GPS-denied indoor flight via ROS 2 offboard control and EKF2 mocap fusion **(video)**
- Developed a robotics research lab from scratch (truss design, OptiTrack camera system procurement) serving 500 students
- Mentored undergraduate teams developing ground robot and Crazyflie platforms for department research projects

MARMot Lab, National University of Singapore — *Singapore*
Research Engineer (Robotics) (Supervisor: Prof. Guillaume Sartoretti) — *Jan 2022 – Aug 2024*
- Deployed a novel gaze-planning (heading optimization) algorithm for legged robot mapping with fixed field-of-view cameras on a real hexapod (ROS, RTAB-Map, RealSense) **(paper)**
- Co-developed an RL-based multi-robot task allocation method, 100x faster than optimization baselines **(ICRA '24 paper)**
- Deployed an inverse kinematics-based controller for omnidirectional locomotion for a hexapod in Python/ROS **(video)**
- Built a motion capture flight arena and a PX4 drone using ESP8266 offboard position control eliminating onboard computer

BITS Goa Mars Rover Team (Kratos), University Rover Challenge — *Goa, India*
Controls Team Lead — *Aug 2020 – Aug 2021*
- Led the controls subsystem, mentoring 10 undergraduate students in robotics software and system integration
- Designed path planning and tracking controllers (PID, Stanley, Pure Pursuit) for autonomous rover navigation
- Developed ROS/C++ drivers and software interfaces integrating sensors and motor drivers with an NVIDIA Jetson

Industry Experience

Komatsu Mining Corp — *Milwaukee, WI*
Robotics Software Intern (Automation) — *Jun 2025 – Aug 2025*
- Developed a startup motor/encoder calibration algorithm in C++ for robotic excavator to eliminate manual motor homing
- Built a C++ teleop interface (steering wheel/pedal) for a PX4 truck by reverse-engineering its RC unit via Arduino + DAC
- Debugged Optitrack motion capture tracking failures by optimizing camera aperture, focus, and calibration parameters
- Contributed to design discussions for tracked autonomous vehicle testing platform integrating LiDAR, camera, and GNSS

Technical Skills

Programming	C, C++ (Modern C++), Python, MATLAB, Bash
Tools & Frameworks	Git, Linux, Docker, CMake, Bazel, **Robotics:** ROS, ROS2, MoveIt!, PX4, RViz, Gazebo, Isaac Sim
Hardware	Raspberry Pi, NUC, Nvidia Jetsons, Arduino, ESP32, STM32, RealSense cameras, Optitrack MoCap
Domain Interests	Robotics Software, System Integration, Motion Planning, Trajectory Optimization, Optimal Control

Teaching / Mentorship Experience

- **Graduate Teaching Assistant** - AEM 3103 Computer Methods in Aerospace Engineering
- **Undergraduate Teaching Assistant** - CS G523 Software for Embedded Systems **(graduate level)**
- **Mentor** - 2026 AEM Summer Research Mentorship Program (SRMP)

Conference Volunteering

- **Student Volunteer** - IEEE Conference on Decision and Control (CDC) 2023 - Singapore
- **Student Volunteer** - IEEE International Conference on Robotics & Automation (ICRA) 2025 - Atlanta, GA, USA

Graduate Coursework

Trajectory Optimization, Optimal Estimation, Robotics, Machine Learning Fundamentals, Deep Learning for Robot Manipulation, Natural Language Processing, Computer Vision

Selected Projects

Optimization-based Motion Planning for 3D Environments (code)

- Built a convex optimization-based collision-free trajectory generation framework in MATLAB using YALMIP/MOSEK
- Modeled cuboidal and cylindrical obstacles as convex sets using superellipse-based formulations for tractable optimization

Cooperative UAV-UGV localization using Extended Kalman Filter (EKF) (code)

- Developed a nonlinear UAV-UGV localization system in MATLAB using an EKF and simulation-based validation
- Evaluated estimator consistency using Monte Carlo testing with NEES/NIS analysis on simulated and real data

Dynamic Visual SLAM for Crowded Indoor Environments

- Integrated ORB-SLAM3 pipeline with Nvidia SegFormer to detect and reject moving objects in crowded environments
- Evaluated in real university environments, achieving 0.077 m ATE and 0.073 m / 1.33° RPE, showing robust localization

Clothbot - Cloth Manipulation using Self Supervised Value Network (poster, web)

- Developed a self-supervised value network policy using spatial action maps for dynamic cloth unfolding on a dual UR5
- Achieved 95% coverage on rectangular cloths and 87.68% on unseen garments (T-shirts) with zero-shot sim-to-real transfer

Flying Ad Hoc Network Simulator for multi-UAV exploration (code)

- Developed a co-simulation platform integrating NS3 and Gazebo through ROS/C++ for testing multi-UAV swarm tasks
- Implemented UAV swarm motion planning (C++) and analyzed network metrics like PDR, hop-by-hop and end-to-end delay
- Simulated a wildfire rescue UAV swarm (PX4 SITL and ROS) for environment surveillance applications **(ACM paper)**